

**NATIONAL UNIVERSITY OF MODERN  
LANGUAGE ISLAMABAD**

**Lab 10**



**Submitted to:**

**Sir Hassan Shah**

**Submitted by:**

**Zulqarnain Hassan**

**Roll No: ..... SP21379**

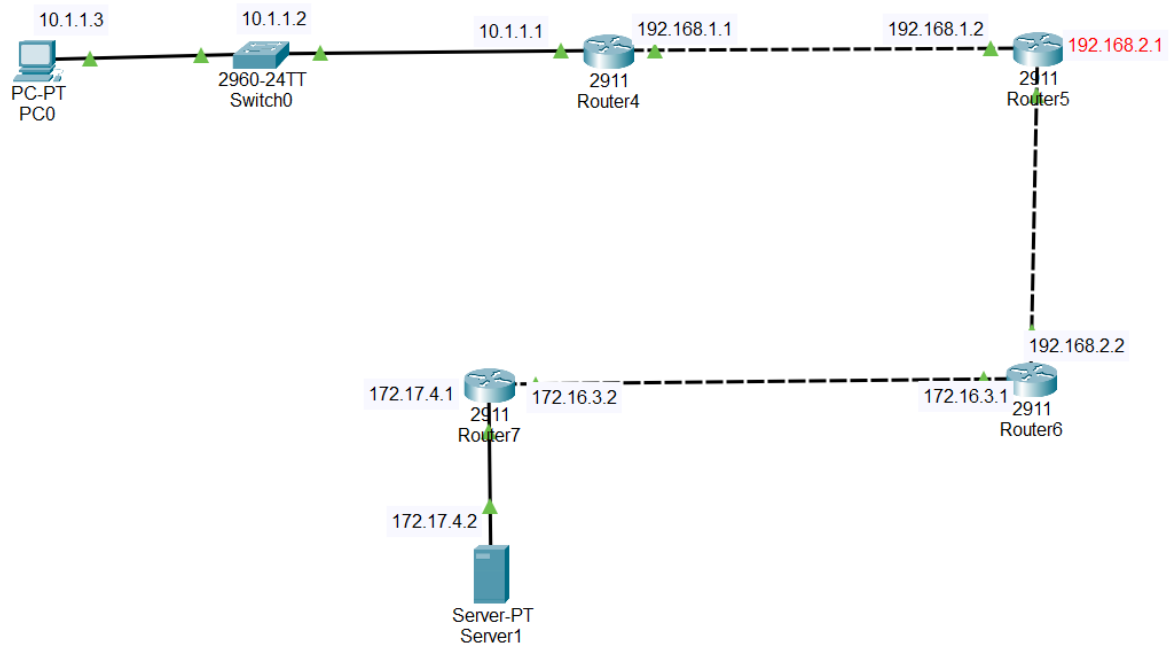
**Subject: ..... computer network**

**Department: ..... BSSE-5-Afternoon**

**Date: ..... 13-12-2024**

## Objectives

- Configure the Router for static routing.



### Task 1

#### Router4:

```
zulqarnain-sp21379-R1(config)#
zulqarnain-sp21379-R1(config)#inter
zulqarnain-sp21379-R1(config)#interface g0/0
zulqarnain-sp21379-R1(config-if)#ip address 192.168.1.1 255.255.255.0
zulqarnain-sp21379-R1(config-if)#exit
zulqarnain-sp21379-R1(config)#interface g0.1
^
% Invalid input detected at '^' marker.

zulqarnain-sp21379-R1(config)#interface g0/1
zulqarnain-sp21379-R1(config-if)#ip address 10.1.1.1 255.0.0.0
zulqarnain-sp21379-R1(config-if)#exi
zulqarnain-sp21379-R1(config)#
zulqarnain-sp21379-R1(config)#
```

## Router5:

```
zulqarnain-sp21379-R2#
zulqarnain-sp21379-R2#
zulqarnain-sp21379-R2#confi
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
zulqarnain-sp21379-R2(config)#
zulqarnain-sp21379-R2(config)#interface g0/0
zulqarnain-sp21379-R2(config-if)#ip address 192.168.1.2 255.255.255.0
zulqarnain-sp21379-R2(config-if)#exit
zulqarnain-sp21379-R2(config)#interface g0/1
zulqarnain-sp21379-R2(config-if)#ip address 192.168.2.1 255.255.255.0
zulqarnain-sp21379-R2(config-if)#exit
zulqarnain-sp21379-R2(config)#
zulqarnain-sp21379-R2(config)#
```

## Router6:

```
zulqarnain-sp21379-R3(config)#
zulqarnain-sp21379-R3(config)#
zulqarnain-sp21379-R3(config)#interface g0/0
zulqarnain-sp21379-R3(config-if)#ip address 192.168.2.2 255.255.255.0
zulqarnain-sp21379-R3(config-if)#exit
zulqarnain-sp21379-R3(config)#interface g0/1
zulqarnain-sp21379-R3(config-if)#ip address 172.16.3.1 255.255.0.0
zulqarnain-sp21379-R3(config-if)#exit
zulqarnain-sp21379-R3(config)#
zulqarnain-sp21379-R3(config)#
```

## Router7:

```
zulqarnain-sp21379-R4(config)#
zulqarnain-sp21379-R4(config)#interface g0/0
zulqarnain-sp21379-R4(config-if)#ip address 172.16.3.2 255.255.0.0
zulqarnain-sp21379-R4(config-if)#exit
zulqarnain-sp21379-R4(config)#
zulqarnain-sp21379-R4(config)#interface g0/1
zulqarnain-sp21379-R4(config-if)#ip address 172.17.4.1 255.255.0.0
zulqarnain-sp21379-R4(config-if)#exit
zulqarnain-sp21379-R4(config)#
zulqarnain-sp21379-R4(config)#
```

## Task 02

### Router4:

```
zulqarnain-sp21379-R1(config)#
zulqarnain-sp21379-R1(config)#
zulqarnain-sp21379-R1(config)#
zulqarnain-sp21379-R1(config)#
zulqarnain-sp21379-R1(config)#ip route 192.168.2.0 255.255.255.0 192.168.1.2
zulqarnain-sp21379-R1(config)#ip route 172.16.0.0 255.255.0.0 192.168.1.2
zulqarnain-sp21379-R1(config)#ip route 172.17.0.0 255.255.0.0 192.168.1.2
zulqarnain-sp21379-R1(config)#
zulqarnain-sp21379-R1(config)#
zulqarnain-sp21379-R1(config)#
```

### Router5:

```
zulqarnain-sp21379-R2(config)#
zulqarnain-sp21379-R2(config)#
zulqarnain-sp21379-R2(config)#
zulqarnain-sp21379-R2(config)#ip route 10.0.0.0 255.0.0.0 192.168.1.1
zulqarnain-sp21379-R2(config)#ip route 172.16.0.0 255.255.0.0 192.168.2.2
zulqarnain-sp21379-R2(config)#ip route 172.17.0.0 255.255.0.0 192.168.2.2
zulqarnain-sp21379-R2(config)#
zulqarnain-sp21379-R2(config)#
zulqarnain-sp21379-R2(config)#
zulqarnain-sp21379-R2(config)#
zulqarnain-sp21379-R2(config)#
```

### Router6:

```
zulqarnain-sp21379-R3(config)#
zulqarnain-sp21379-R3(config)#
zulqarnain-sp21379-R3(config)#ip route 10.0.0.0 255.0.0.0 192.168.2.1
zulqarnain-sp21379-R3(config)#ip route 192.168.1.0 255.255.255.0 192.168.2.1
zulqarnain-sp21379-R3(config)#ip route 172.17.0.0 255.255.0.0 172.16.3.2
zulqarnain-sp21379-R3(config)#
zulqarnain-sp21379-R3(config)#
zulqarnain-sp21379-R3(config)#
zulqarnain-sp21379-R3(config)#
zulqarnain-sp21379-R3(config)#
```

### Router7:

```
!----- Configuration Commands, one per router, and their output -----
zulqarnain-sp21379-R4(config)#
zulqarnain-sp21379-R4(config)#
zulqarnain-sp21379-R4(config)#ip route 192.168.2.0 255.255.255.0 172.16.3.1
zulqarnain-sp21379-R4(config)#ip route 192.168.1.0 255.255.255.0 172.16.3.1
zulqarnain-sp21379-R4(config)#ip route 10.0.0.0 255.0.0.0 172.16.3.1
zulqarnain-sp21379-R4(config)#
zulqarnain-sp21379-R4(config)#
zulqarnain-sp21379-R4(config)#
```

### Task 3

```
zulqarnain-sp21379-R3>en
zulqarnain-sp21379-R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, GigabitEthernet0/1
L       172.16.3.1/32 is directly connected, GigabitEthernet0/1
S       172.17.0.0/16 [1/0] via 172.16.3.2
S       192.168.1.0/24 [1/0] via 192.168.2.1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.2/32 is directly connected, GigabitEthernet0/0

zulqarnain-sp21379-R3#
```

### Ping index.html from Pc0

```
~.~
C:\>ping index.html

Pinging 172.17.4.2 with 32 bytes of data:

Reply from 172.17.4.2: bytes=32 time=10ms TTL=124
Reply from 172.17.4.2: bytes=32 time=1ms TTL=124
Reply from 172.17.4.2: bytes=32 time<1ms TTL=124
Reply from 172.17.4.2: bytes=32 time=1ms TTL=124

Ping statistics for 172.17.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```